
```
def overlapping_sums(the_list, sum_length):
    length_of_list = len(the_list)
    the_list.extend(the_list)
    output = []
    for n in range(0, length_of_list):
        output.append(sum(the_list[n:(n + sum_length)]))
    return(output)
```

We can run this test on a new random list like so:

```
import matplotlib.pyplot as plt
overlap = overlapping_sums(list_random(211111, 111112, 300007), 12)
plt.hist(overlap, 20, facecolor = 'blue', alpha = 0.5)
plt.title('Results of the Overlapping Sums Test')
plt.xlabel('Sum of Elements of Overlapping Consecutive Sections of List')
plt.ylabel('Frequency of Sum')
plt.show()
```

We created a new random list by running `list_random(211111, 111112, 300007)`. This new random list is long enough to make the overlapping sums test perform well. The output of this code is a histogram that records the frequency of the observed sums. If the list resembles a truly random collection, we expect some of the sums to be high and some to be low, but we expect most of them to be near the middle of the possible range of values. This is exactly what we see in the plot output ([Figure 5-2](#)).